Rudray A. Dave

Phone: +91-9429876725| **Email:** rudraydave@gmail.com
LinkedIn | LeetCode | GitHub | Portfolio

Education

B.Tech in Artificial Intelligence 2023 – Present
Sardar Vallabhbhai National Institute of Technology, Surat CGPA: 8.55 / 10 | View Marksheet

AI Honours Programme (NIT Surat) 2024 – Present
Advanced Algorithms, Generative AI, Advanced Deep Learning, Nature-Inspired Computing CGPA: 9.75 / 10

Work Experience

R&D Intern August 2026 – Present
Atomberg Technologies

- Worked on AI-driven workflows to extract structured knowledge from P&ID engineering diagrams and automate the placement of Leviton electrical components, streamline engineering documentation review and improve workflow efficiency.

AI Workflow Automation Intern May 2026 – July 2026
L&T Technology Services Intern Certificate

- Worked on AI-driven workflows to extract structured knowledge from P&ID engineering diagrams and automate the placement of Leviton electrical components, streamline engineering documentation review and improve workflow efficiency.

Gujarat Police Project Intern Jan 2026 – April 2026
Applied Machine Learning and Wearable Sensor Analytics

- Developed machine learning pipelines to infer cognitive and operational states such as stress, focus, and activity from wearable physiological signals, tackling noisy real-world sensor data through feature engineering and signal processing.

AI Research Intern Oct 2025 – Dec 2025
Applied Research in Artificial Intelligence under Dr. Rahul Dixit Letter of Recommendation

- Trained and evaluated three PyTorch CNN architectures on spectrogram representations for speech forgery detection, and co-authored two peer-reviewed research articles investigating model robustness across clean and degraded audio conditions.

Technical Projects

Inference-Time Learning via Persistent Neural Memory 2026
PyTorch, Continual Learning, RL, LLM Agents GitHub

- Developed and benchmarked a neural memory architecture for online adaptation with a frozen base LLM, achieving 45.5% recovery of previously failed tasks vs. 12.5% for a memoryless baseline, while evaluating continual learning, memory consolidation, and stability.

Financial Fraud Detection using Distributed ML & Cluster 2026
Big Data Analytics Course Project GitHub

- Built a distributed fraud-detection pipeline over 6.3M+ PaySim transactions using Hadoop, Spark, Hive, and PySpark; implemented a 2-layer LSTM with per-account sliding windows, achieving 0.9631 F1-score and 0.9996 ROC-AUC.

Cortogen - SaaS 2026
Chrome Extension, LLM Orchestration Website

- Built and deployed a production AI Chrome extension with persistent cross-session LLM memory using FastAPI, LangChain, FAISS, PostgreSQL, and AWS to reduce manual effort and token usage.

Audio Forgery Detection & LibriSpeech Copy-Move Dataset 2025
Audio Forensics & Deep Learning, Dataset Accepted in Springer Conference Proceedings Certificate

- Created a 57K+ sample audio forgery benchmark and trained PyTorch CNN models achieving a 12% reduction in false positives over baseline methods.

Ongoing Research

Compute-Efficient LLM Training via Adaptive Data Selection 2026 – Present
Under the guidance of Dr. Sudhakar Mishra GitHub

- Evaluating adaptive data selection for compute-efficient LLM reasoning through controlled, matched-GPU-budget experiments comparing selection overhead, training efficiency, and MATH-500 accuracy against random sampling across multiple random seeds.

Hackathons & Competitions

Google Winter of Code February 2025
Bindi's Cupcakery Website | GitHub

- Built and deployed a cloud-kitchen bakery website using the PERN stack, implementing secure authentication, real-time inventory tracking, and order management; secured **1st place** in Google Winter of Code.

Odoo Hackathon August 2025
QuickCourt – Court Booking and Management Platform Video | GitHub

- Developed QuickCourt with a team in a 24-hour hackathon, building a court booking platform with availability scheduling, authentication, booking confirmations, payment processing, and role-based access; the project was selected as a **finalist**.

Kaggle Datathon October – November 2025
Google Developer Groups on Campus GitHub

- Worked with a team to develop a machine learning solution for the Kaggle Datathon, securing **11th rank among hundreds of participants** in the competition held from October 31 to November 2, 2025.

DotSlash 9.0 March 2026
CivicAssist – Government Scheme Discovery Platform GitHub

- Helped develop CivicAssist, an offline-first platform for discovering government schemes, with eligibility assessment, personalized recommendations, and AI-powered policy simplification using Next.js, Express, Supabase, and Gemini.

Skills

AI and LLM Systems:RAG, Prompt Engineering, Vector Databases, Large Language Models, Model Fine-Tuning, LoRA, LangChain, Agentic Systems

Machine Learning: PyTorch, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Hyperparameter Tuning

Web Development: Next.js, RESTful APIs, MERN, PERN, MongoDB, GraphDB, Git, GitHub, Flask, FastAPI

Big Data and DevOps: Apache Hadoop, Apache Spark, PySpark, HDFS, YARN, Hive, Docker, AWS

Programming Languages:C, Python, SQL

Relevant Coursework

Mathematics & Algorithms: Linear Algebra and Statistics, Discrete Mathematics, Fundamentals of Engineering Mathematics, Design and Analysis of Algorithms, Advanced Algorithms

Artificial Intelligence & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Generative AI, Natural Language Processing

Computing & Data Science: Data Science, Database Management Systems, Big Data Analysis and Visualization, Cloud Computing, Distributed Computing

Computer Systems: Computer Organization, Operating Systems, Computer Networks, Digital Electronics and Logic Design, Microprocessor and Interfacing Techniques

Specialized Topics: Nature-Inspired Computing and Optimization, Information Security and Cryptography

Positions of Responsibility

Career Development Cell Coordinator – Department of Artificial Intelligence, SVNIT Present
Branch Counsellor – Department of Artificial Intelligence, Student Affairs Council, SVNIT 2025 – 2026
Mentor – Google Winter of Code, GDGC SVNIT 2025 – 2026